



UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES
BSc Degree First Year First Semester Course Unit Examination – Sep./Oct. 2024

699

DEPARTMENT OF ENGLISH LANGUAGE TEACHING

ENG 101 2.0 Compulsory English (Scientific Communication)

Time: Two (02) hours
Answer ALL questions

No. of questions: 08

No. of pages: 07

Total marks: 100

Index Number:

1. Select the most suitable word from the box given below and fill in the blanks. There is one extra word.

survival	pollination
integral	habitat
species	equilibrium

Flora and fauna are essential components of an ecosystem. They play a critical role in maintaining a balanced and healthy environment. The importance of flora or plants, lies in their ability to produce oxygen through photosynthesis which provides food and (1).....for various species, and contribute to soil formation and nutrient cycling. Fauna, or animals, are also vital to the ecosystem as they help in seed dispersal, (2).....and maintaining the food chain.

Plants are the primary producers in an ecosystem, converting sunlight, water and carbon dioxide into organic matter. This process, known as photosynthesis, produces oxygen as a byproduct, which is essential for the (3).....of all aerobic organisms. Fauna, on the other hand, plays a crucial role in maintaining the delicate (4).....of an ecosystem. Animals help in seed dispersal, ensuring that plant species can spread and colonize new areas. Animals are an (5).....part of the food chain, serving as both predators and prey, which helps to regulate population sizes and control the spread of diseases. (05 marks)

2. Construct questions to get the words in bold print as answers.

i) My uncle and aunt went to New Zealand **last December**.

.....?

ii) Sampson visits his grandparents **thrice a month**.

.....?

iii) Mary's sister has studied **in Michigan university**.

.....?

iv) I couldn't submit the assignment on time because I was down with flu.

.....?

v) The girls went on shopping with their best friends.

.....?

(05 marks)

3. Fill in the blanks with the correct word given in the brackets.

There is a famous expression in English: "Stop the world, I want to get off!"

(i).....(Those/This/These) expression refers to a feeling of panic, or stress, that makes

(ii).....(the/a/an) person want to stop whatever they are doing, try to relax, and become calm again. 'Stress' means pressure or tension. It is one of the

(iii).....(commonest/most common/commoner) causes of health problems in modern life. Too much stress results in physical, emotional, and mental health problems. There are numerous physical effects of stress. Stress can affect the heart. It can increase the pulse rate, make the heart miss beats, and can cause high blood pressure. Stress can affect (iv).....(the/a/an)

respiratory system. It can lead to asthma. It can cause a person to breathe too fast, resulting in a loss of important carbon dioxide. Stress can affect the stomach. It can cause stomach aches and problems digesting food. These are only (v).....(few/a few/fewer) examples of the wide

range of illnesses and symptoms resulting from stress. Emotions are also (vi).....(easy/easily/easier) affected by stress. People suffering from stress often feel

anxious. They may have panic attacks. They may feel tired all the time. When people are under stress, they often overreact to (vii).....(a little/little/least) problems. For example, a

normally gentle parent under (viii).....(a lot/lot/a lots) of stress at work may yell at a child for dropping a glass of juice. Stress can make people angry, moody, or nervous.

Long-term stress can lead to a variety of serious mental illnesses. Depression, (ix).....(an/a/the) extreme feeling of sadness and hopelessness, can be the result of continued and increasing stress. Alcoholism and other addictions often develop as a result of overuse of alcohol or drugs to try to

relieve stress. Eating disorders, such as anorexia, are sometimes caused by stress and are often made worse by stress. If stress is allowed to continue, then one's mental health is put at risk. It is obvious that stress is a serious problem. It attacks the body. It affects the emotions. Untreated, it may eventually result in mental illness. Stress has a great influence on the health and well-being of

(x).....(we, our, us) bodies, our feelings, and our minds. So, reduce stress: stop the world and rest for a while.

(10 marks)

4. Complete the following sentences meaningfully.

i) I got late to the lectures because.....

ii) Mary and Peter submitted the project report on time but.....

iii) When.....my mother was preparing our meals.

iv) Although my sister studied hard.....

v) Mrs. Peterson didn't telephone me nor.....

vi) Which dress do you prefer? The blue one over there or.....

- vii) I forgot to bring my pen drive yet.....
- viii) Even though....., nobody wanted to listen to her.
- ix) Since Eliza couldn't go to work yesterday,
- x) It was very cold outside so.....

(10 marks)

5. (A) Choose the correct form of the verbs in brackets.

Fiat, the automobile company (i).....(has started/started/was started) by a group of Italian businessmen in 1899. In 1903, Fiat, (ii).....(was produced/produced/produces) 132 cars. Some of these cars (iii).....(were exporting/exported/were exported) by the company to the United States and Britain. In 1920, Fiat (iv).....(was being started/started/ was started) making cars in a new factory at Lingotto, near Turin. There was a track on the roof where the cars (v).....(are being tested/tested/were tested) by technicians. In 1936, Fiat launched the Fiat 500. This car (vi).....(called/was called/has called) the Topolino- the Italian name for Mickey Mouse. The company grew and in 1963 Fiat (vii).....(had been exported/exported/was exported) more than 300,000 vehicles. Today, Fiat is based in Turin and its cars (viii).....(sold/have sold/are sold) all over the world.

(B) Rewrite the following passage in the passive form. See the example given below.

My uncle painted this picture. He has done several paintings like this. Mostly, he has painted beautiful natural sceneries. Everybody appreciates his efforts very much. He does a lot of hard work at the completion of these paintings. He organized an exhibition to exhibit his most exciting paintings. At the exhibition, Mr. Green has offered him a lot of money for one of his paintings. So, my uncle will deliver the painting tomorrow to Mr. Green.

This picture was painted by my uncle.....

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(15 marks)

6. Make meaningful sentences using the following words. You may use the verbs in any tense. Do not change the form of the nouns, adjectives and adverbs.

development	organize	annually	operation
skillful	contamination	scientific	educate
efficient	establish	analyze	compulsory
nutrition	carefully	ethics	

- i)
- ii)
- iii)
- iv)
- v)
- vi)
- vii)
- viii)
- ix)
- x)
- xi)
- xii)
- xiii)
- xiv)
- xv)

(15 marks)

7. Read the following text and answer the questions given below. (20 marks)

At the intersection of land and sea, mangrove forests support a wealth of life, from starfish to people, and may be more important to the health of the planet than we ever realized.

- A) Mangroves live life on the edge. With one foot on land and one in the sea, these botanical amphibians occupy a zone of desiccating heat, choking mud, and salt levels that would kill an ordinary plant within hours. Yet the forests mangroves form are among the most productive and biologically complex ecosystems on Earth. Birds roost in the canopy, shellfish attach themselves to the roots, and snakes and crocodiles come to hunt. Mangroves provide nursery grounds for fish; a food source for monkeys, deer, tree-climbing crabs, even kangaroos; and a nectar source for bats and honeybees.
- B) As a group, mangroves can't be defined too closely. There are some 70 species from two dozen families- among them palm, hibiscus, holly, plumbago, acanthus, legumes and myrtle. They range from prostrate shrubs to 200-foot-high (60 meters) timber trees. Though most prolific in Southeast Asia, where they are thought to have originated, mangroves circle the globe. Most live within 30 degrees of the Equator, but a few hardy types have adapted to temperate climates, and one lives as far from the tropical sun as New Zealand. Wherever they live, they share one thing in common: They're brilliant adapters. Each mangrove has

an ultrafiltration system to keep much of the salt out and a complex root system that allows it to survive in the intertidal zone. Some have snorkel-like roots called pneumatophores that stick out of the mud to help them take in air; others use prop roots or buttresses to keep their trunks upright in the soft sediments at tide's edge.

C) These plants are also land builders par excellence. Some Aborigines in Northern Australia believe one mangrove species resembles their primal ancestor, Giyapara, who walked across the mudflats and brought the tree into existence. The plants' interlocking roots stop river-borne sediments from coursing out to sea, and their trunks and branches serve as a palisade that diminishes the erosive power of waves.

D) Despite their strategic importance, mangroves, worldwide are under a huge threat. They are sacrificed for salt pans, aquaculture ponds, housing developments, roads, port facilities, hotels, golf courses and farms. They also die from a thousand indirect cuts: oil spills, chemical pollution, sediment overload and disruption of their sensitive water and salinity balance. Calls for mangrove conservation gained a brief but significant hearing following the 2004 Indian Ocean Tsunami where mangrove forests were intact, they served as natural breakwaters, dissipating the energy of the waves, mitigating property damage, perhaps saving lives.

E) Throughout the tropical world; despite the importance of mangrove forests; they are being destroyed. One of the greatest threats to mangrove survival comes from shrimp farming. At first glance, shrimp might seem the perfect export for a poor country in a hot climate. Rich countries have an insatiable appetite for it (shrimp has overtaken tuna to become America's favorite seafood), and the developing world has the available land and right climate to farm it.

F) As serious as the threat from shrimp farming is to the world's remaining mangroves, there looms a potentially more disastrous problem: rising sea levels. Standing as they do at the land's frontiers, mangroves will be the first terrestrial forests to face the encroaching tides.

G) Loss of mangrove forests could prove catastrophic in ways only now becoming apparent. For more than 25 years Jin Eong Ong, a retired professor of marine and coastal studies in Penang, Malaysia has been exploring a less obvious mangrove contribution: What role might these forests play in climate change? Ong and his colleagues have been studying the carbon budget of mangroves-the balance sheet that compares all the carbon inputs and outputs of the mangrove ecosystem-and they've found that these forests are highly effective carbon sinks. They absorb carbon dioxide, taking carbon out of circulation and reducing the amount of greenhouse gas. Mangroves may have the highest net productivity of carbon of any natural ecosystem, and as much as a third of this may be exported in the form of organic compounds to mudflats. Mangroves, it seems, are carbon factories, and their demolition robs the marine environment of a vital element. Ong's team has also shown that a significant portion of the carbon ends up in forest sediments, remaining sequestered there for thousands of years. Conversion of a mangrove forest to a shrimp pond changes a carbon sink into a carbon source, liberating the accumulated carbon back into the atmosphere but 50 times faster than it was sequestered. According to Eong Ong, if mangroves were to become recognized as carbon-storage assets, then it could radically alter the way these forests are valued.

A) Which paragraph contains the following information? Write the appropriate letter within brackets.

- i) The types of climatic conditions in which mangroves are grown and survived. ()
- ii) Mangroves provide a rich ecosystem for the survival of various living beings. ()
- iii) Shrimp farming can be considered as a major danger to mangrove forests. ()

- iv) Mangroves play a vital role in absorbing carbon dioxide. ()
v) Mangroves protect the land from intruding tidal waves. () (05 marks)

B) Say whether the following statements are TRUE/FALSE/NOT GIVEN.

- i) Mangroves hardly provide a nectar source for bats and honeybees.
ii) It is believed that mangroves are originated in Southeast Asia.
iii) The trunks and branches of mangrove plants reduces sea erosion.
iv) Mangroves highly supports the biodiversity and the equilibrium in the environment.

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v) Converting a mangrove forest into a shrimp pond enhances the absorption of Carbon in the atmosphere. (05 marks)

C) Answer the following questions.

- i) According to the text, why do you think mangroves are brilliant adapters?

.....
.....

- ii) Why do mangroves are important for animals?

.....
.....

- iii) Name four categories of mangrove species?

.....
.....

- iv) What are the causes of the destruction of mangrove forests?

.....
.....

- v) According to the text, explain the importance of conserving mangrove forests?

.....
.....
.....

(10 marks)

71 7
8. Write an essay on the topic given below. Word limit 150-200

(20 marks)

"Should universities and other educational institutions implement strict rules and regulations on junk food sales in their canteens?" What is your opinion about it? Justify your view by using relevant examples.
